



RUTGERS-NEW BRUNSWICK

School of Engineering

Department of Industrial and Systems Engineering

COURSE PROJECT

KAGGLE PREDICTION COMPETITION

PREDICTING LOAN PAYBACK

BY: ABDELRAHMAN SAWY

TO: PROF. MYONG K JEONG

694 – ADVANCED TOPICS IN I.E. – DATA MINING I

AGENDA

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COMPETITION OVERVIEW

KAGGLE – Playground Prediction Competition ([Predicting Loan Payback | Kaggle](#))

Playground Series – Season 5, Episode 11 (Start Date: November 1, 2025; Final Deadline: November 30, 2025)

Goal: Predict the probability that a borrower will pay back their loan.

Datasets Description: The dataset for this competition (both train and test) was generated from a deep learning model trained on the [Loan Prediction dataset](#). Feature distributions are close to, but not the same, as the original.

Original dataset (loan_dataset_20000.csv; 20,000 observations, 21 features, no missing values)

Competition:

Training dataset (train.csv; 593,993 observations, 11 features, no missing values)

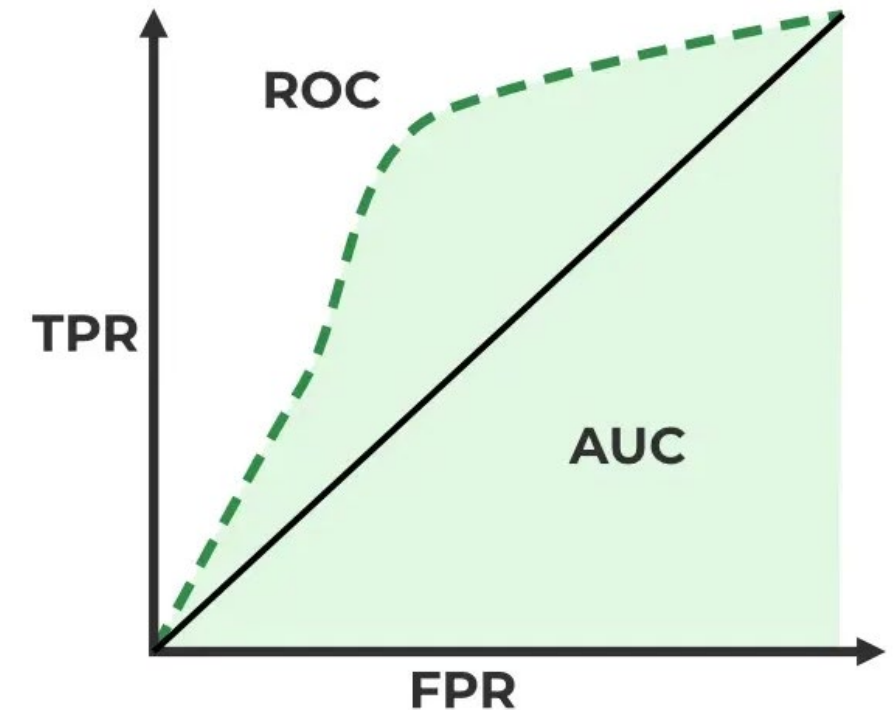
Testing dataset (test.csv; from id 593,994 to id 848,562, no missing values)

To explore the model performance. I will use the original dataset along with the competition dataset to analyze the differences between them and whether incorporating the original dataset in training improves performance.

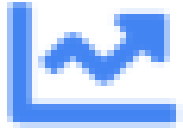
COMPETITION OVERVIEW

About the evaluation metric – AUC:

- Kaggle scores submissions by AUC, the area under the ROC (Receiver Operating Characteristic) curve, computed from the predicted probability vs. the true 0/1 target.
- ROC curve: Plots True Positive Rate (TPR), aka sensitivity, vs. False Positive Rate (FPR), aka specificity, $1 - \text{TPR}$, over all classification thresholds. It represents the tradeoff between the sensitivity and specificity of a classifier.
- AUC: Measures the area under the ROC curve.
 - High AUC (close to 1): Model distinguishes between positive and negative classes.
 - Low AUC (close to 0): Model struggles to differentiate between the two classes.
 - AUC around 0.5: Model doing random guessing (doesn't learn any patterns).



COMPETITION OVERVIEW



Problem Statement

Predict the probability that a borrower will pay back their loan.

Supervised binary classification problem.

$$\hat{p}(\mathbf{x}) \approx P(y = 1|\mathbf{x})$$



Original Dataset

21 features

1 target variable

Each observation represents a loan issued to a borrower, described by 21 borrower and loan features.



Competition Dataset

11 features

1 target variable

Generated from deep learning model trained on original dataset.



Goal

Build a probabilistic model that estimates the likelihood a borrower will fully repay their loan.

Enable banks to make better loan approval and risk-based pricing decisions.

DATASETS DESCRIPTION

Original dataset (full features)
Competition dataset (highlighted)

	Input Variable (Feature)	Type	Description
Demographics	age	int64	Borrowers age in years
	gender	category	Male, Female, Other
	marital_status	category	Single, Married, Divorced, Widowed
	education_level	category	High School, Bachelor's, Master's, PhD, Other
Financial Information	annual_income	float64	Borrower's yearly income
	monthly_income	float64	Borrower's monthly income
	employment_status	category	Employed, Self-employed, Unemployed, Student, Retired
	debt_to_income_ratio	float64	Ratio of borrower's debt to their income → Lower = Better
	credit_score	int64	Credit bureau score (FICO) → Higher = Less risky
Loan Information	loan_amount	float64	Amount of loan taken
	loan_purpose	category	Business, Car, Debt consolidation, Education, Home, Medical, Vacation, Other
	interest_rate	float64	Loan per annual interest rate (%)
	loan_term	int64	Loan repayment duration (months, e.g., 36 or 60)
	installment	float64	Monthly installment
	grade_subgrade	category	Risk category assigned to loan (A1, ..., A5, B1, ..., B5, C1, ..., C5, D1, ..., D5, E1, ..., E5, F1, ..., F5)

	Input Variable (Feature)	Type	Description
Credit History	num_of_open_accounts	int64	Total active credit accounts
	total_credit_limit	float64	Borrower's total available credit limit
	current_balance	float64	Borrower's outstanding balance (loan + credit card)
	delinquency_history	int64	Count of late payments in borrower's history
	public_records	int64	Negative public records (e.g., bankruptcies, legal actions)
	num_of_delinquencies	int64	Total delinquencies (missed payments)

Target Variable	Type	Description
loan_paid_back	int64	1 → Borrower paid loan in full 0 → Borrower defaulted (didn't repay fully)

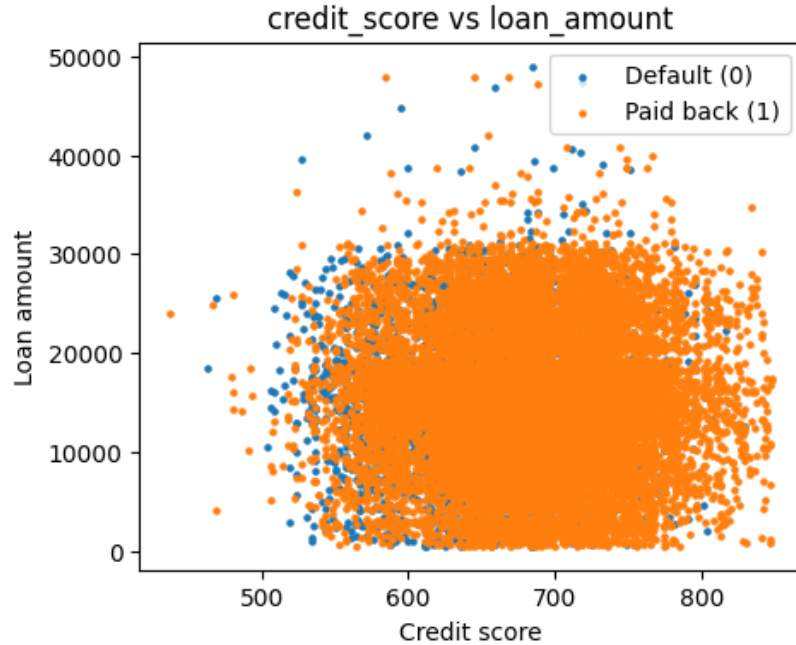
DATASETS DESCRIPTION

Competition dataset – sample observations

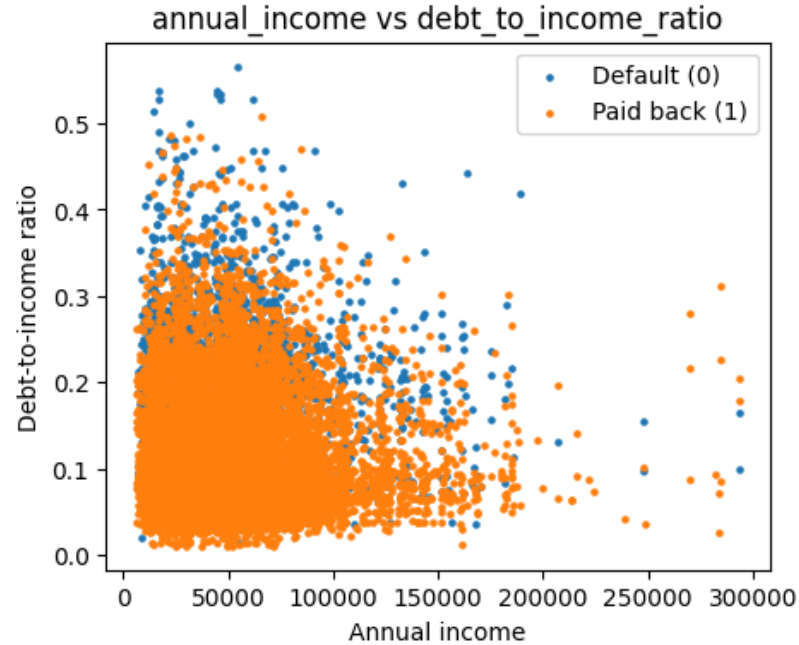
Training Dataset				
id	Input Variables			Target Variable
	annual_income	debt_to_income_ratio	credit_score	loan_paid_back
0	29367.99	0.084	736	1
	loan_amount	interest_rate	gender	
	2528.42	13.67	Female	
	marital_status	education_level	employment_status	
	Single	High School	Self-employed	
	loan_purpose	grade_subgrade		
	Other	C3		
⋮				
id	annual_income	debt_to_income_ratio	credit_score	loan_paid_back
593993	48959.52	0.096	752	1
	loan_amount	interest_rate	gender	
	7707.73	10.31	Male	
	marital_status	education_level	employment_status	
	Married	High School	Employed	
	loan_purpose	grade_subgrade		
	Education	B3		

Testing Dataset				
id	Input Variables			Target Variable
	annual_income	debt_to_income_ratio	credit_score	loan_paid_back
593994	28781.05	0.049	626	?
	loan_amount	interest_rate	gender	
	11461.42	14.73	Female	
	marital_status	education_level	employment_status	
	Single	High School	Employed	
	loan_purpose	grade_subgrade		
	Other	D5		
⋮				
id	annual_income	debt_to_income_ratio	credit_score	loan_paid_back
848562	45627.53	0.118	624	?
	loan_amount	interest_rate	gender	
	19246.14	11.64	Female	
	marital_status	education_level	employment_status	
	Married	High School	Employed	
	loan_purpose	grade_subgrade		
	Car	D3		

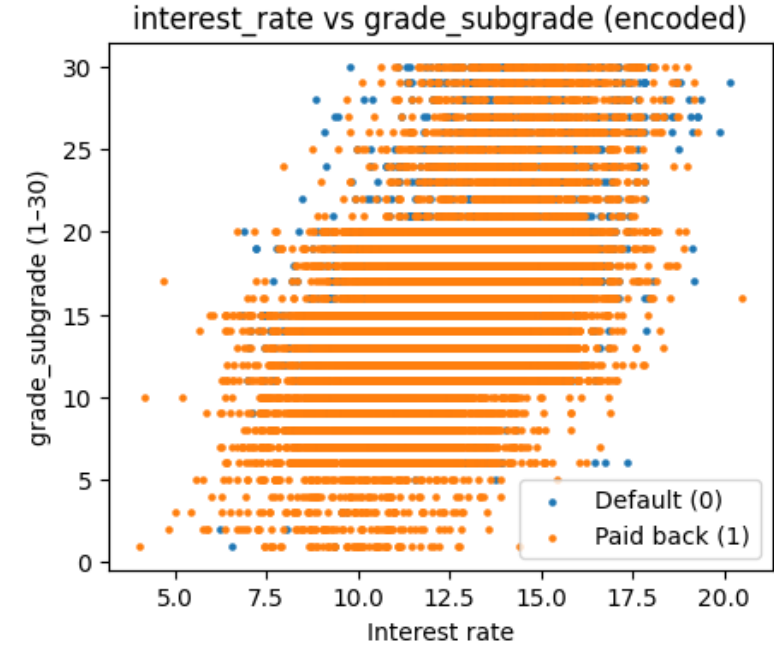
EDA



Two classes largely overlap. Defaults occur more at lower credit score and higher loan amounts.



Paid back concentrated at lower ratio, while defaults are more frequent when ratio is high.



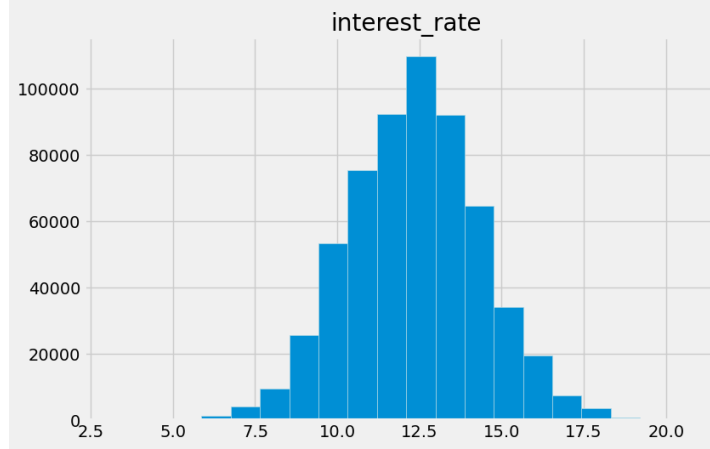
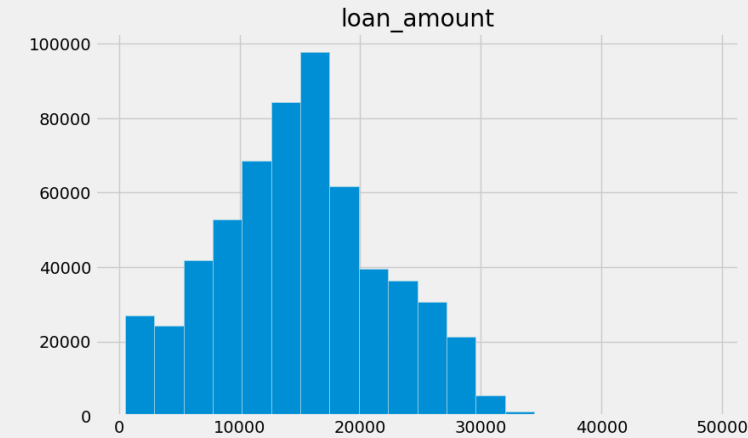
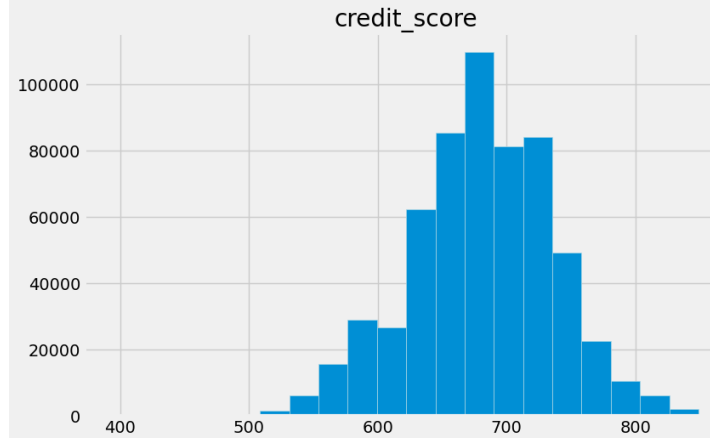
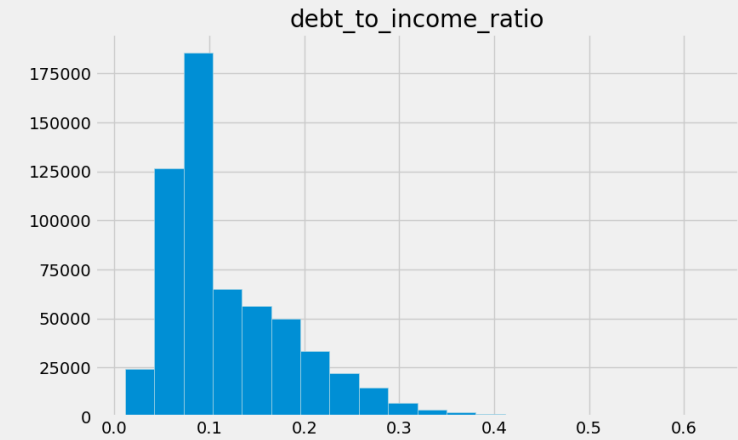
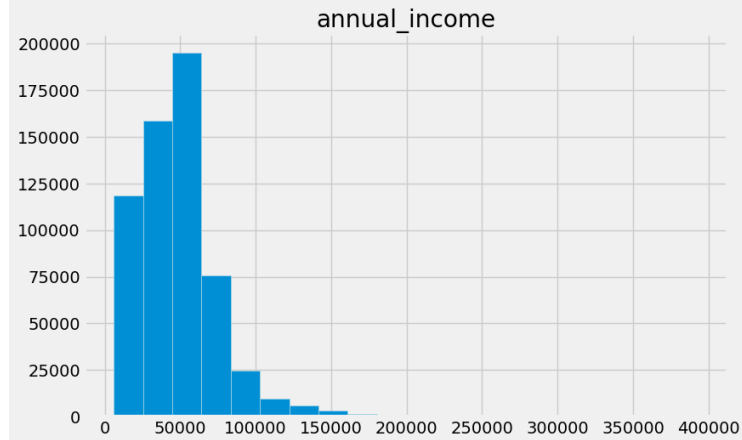
Upward trend: Riskier grades/subgrades receive higher interest rates. Defaults are more common in these high rate, high grade number bands.

EDA

Annual income and debt to income ratio are strongly right-skewed.

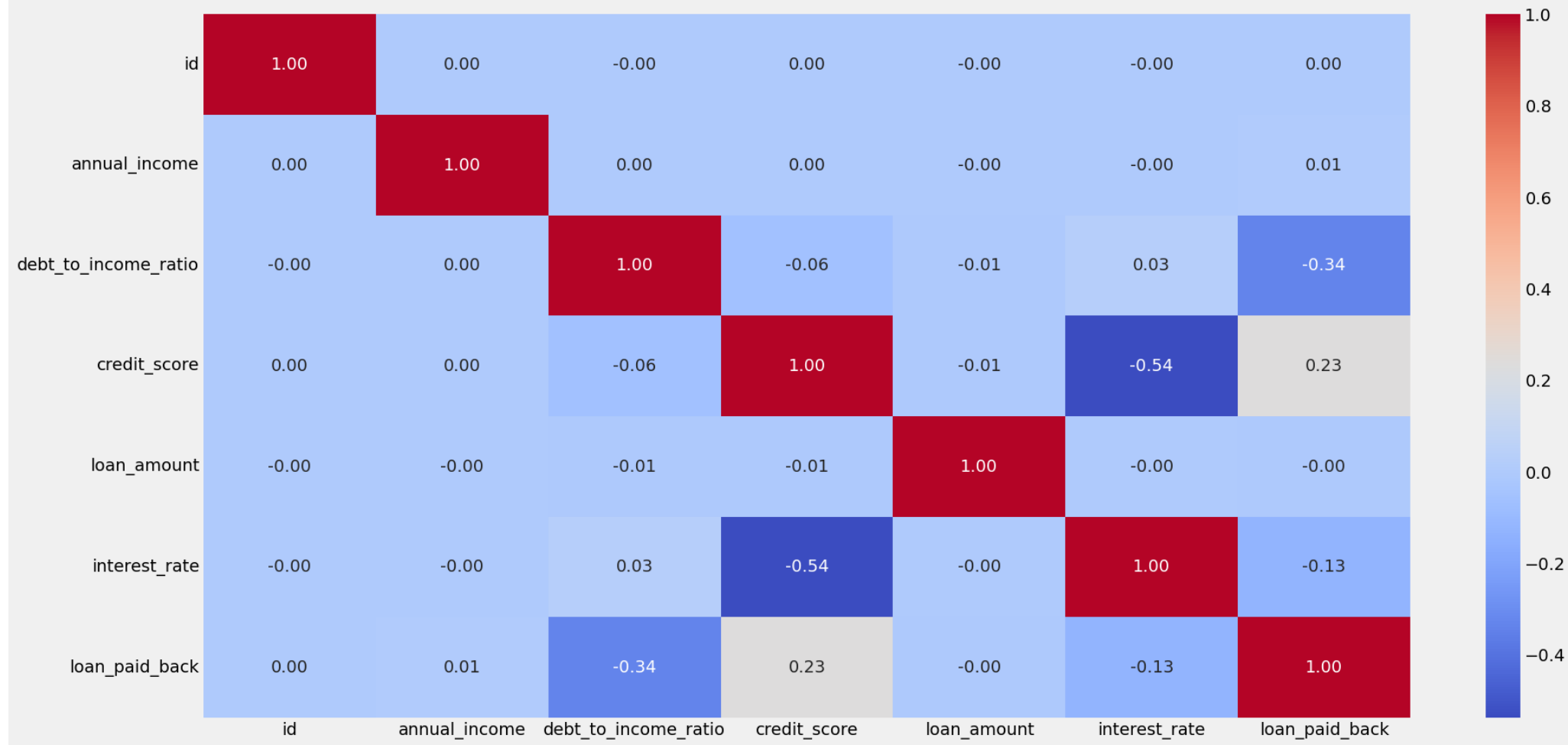
Credit score, loan amount, and interest rate are closer to well behaved normal distributions.

Simple scaling and preprocessing of categorical features should give a strong baseline model.



EDA

Numerical features



Strongest relationships with loan_paid_back:

debt_to_income_ratio (negative correlation)

credit_score (positive correlation)

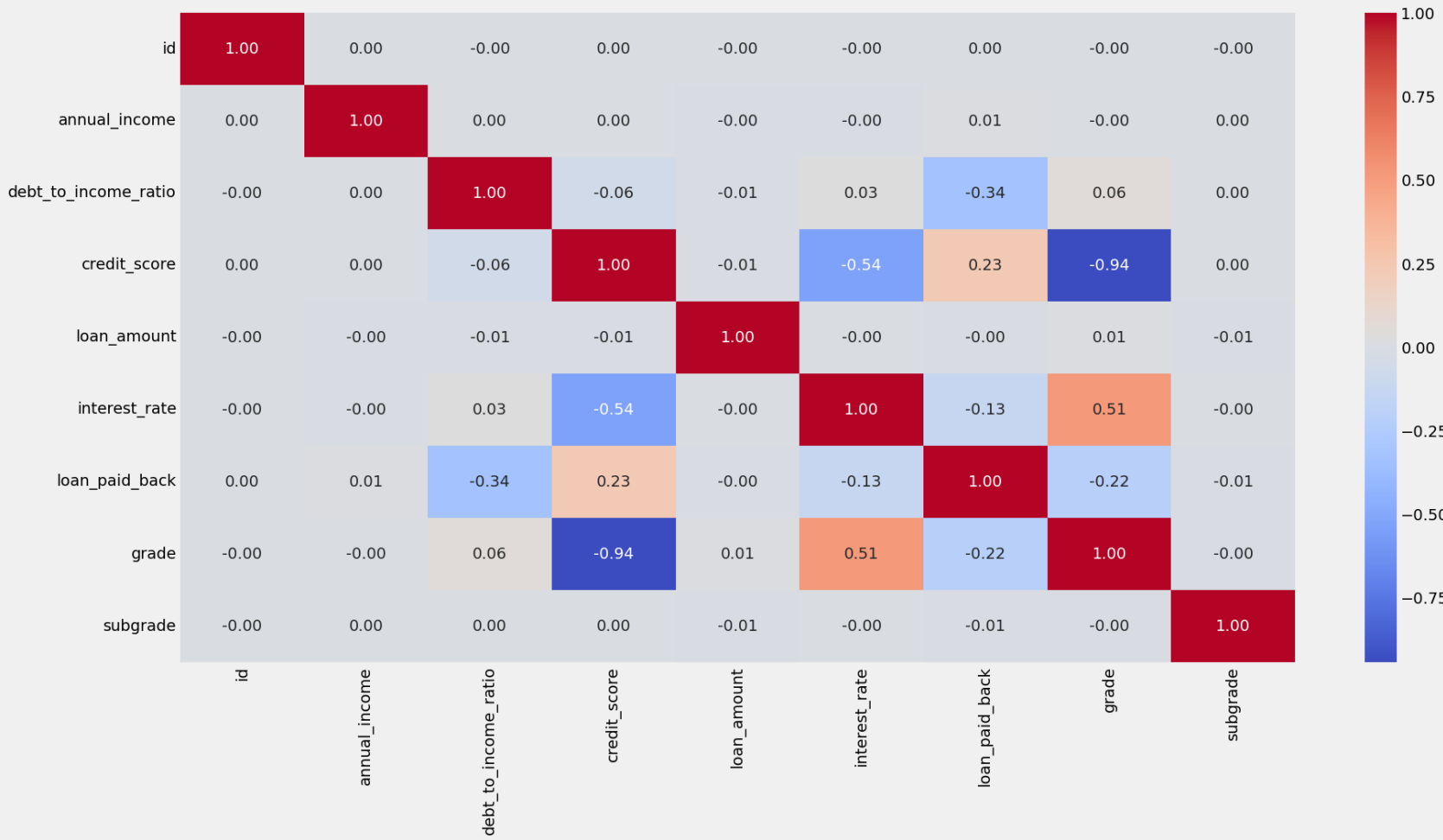
interest_rate (negative correlation) with credit_score & loan_paid_back

→ Higher debt = Lower repayment probability

→ Higher score = Higher repayment probability

→ Higher risk borrowers charged with higher rates and being more likely to default

Keep all features but place emphasis on capturing the nonlinear effects of debt-to-income, credit score, and interest rate.



PREPROCESSING

id: 593994

annual_income: 119728

debt_to_income_ratio: 526

credit_score: 399

loan_amount: 111570

interest_rate: 1454

gender: 3

marital_status: 4

education_level: 5

employment_status: 5

loan_purpose: 8

loan_paid_back: 2

grade: 6

subgrade: 5

Separating grade into 6 levels and subgrade into 5 levels

→ Reduce 30-level category (A1-F5) into 2 structured features

→ Less sparse model than treating each subgrade as a separate category

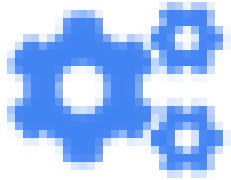
grade → { negative correlation with credit score
positive correlation with interest rate
negative correlation with loan paid back
safer borrowers get better grades and lower rate
worse grades linked to higher default risk

Subgrade → uncorrelated with others; adds linear information



MODEL 1 (BASELINE)

LIGHTGBM + K-FOLD CV



Methodology

LGBM Classifier

K-Fold with $K = 10$, scored by AUC

Gradient-boosted decision trees handles non-linearities, correlations, and mixed features types.



Mathematical view

Learn additive ensemble of trees:

$$f(\mathbf{x}) = \sum_{m=1}^M \eta_m h_m(\mathbf{x})$$

Outputs probability of repayment via logistic link

$$\hat{p}(\mathbf{x}) = P(y = 1|\mathbf{x}) = \sigma(f(\mathbf{x})) = \frac{1}{1 + e^{-f(\mathbf{x})}}$$

Training minimizes logistic loss

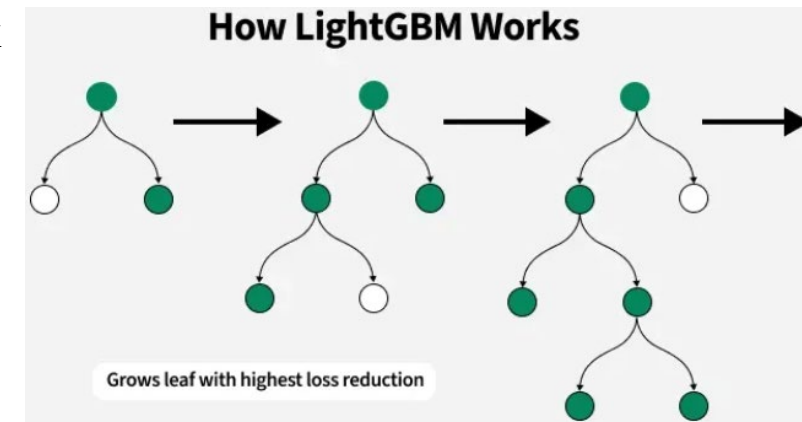
$$\mathcal{L} = - \sum_{i=1}^n [y_i \log \hat{p}(\mathbf{x}_i) + (1 - y_i) \log(1 - \hat{p}(\mathbf{x}_i))]$$



Outcome

Mean ROC-AUC across CV folds:
0.9198752

Kaggle score: 0.91972



FEATURE IMPORTANCE ANALYSIS (MODEL 1)

Most influential features are debt to income ratio, credit score, loan amount, annual income, and interest rate, followed by employment status and grade/subgrade encoding.

Results confirm that both ability to repay (income, DTI ratio, employment) and credit (score, risk, rate) affect the model's decision most.

Use this ranking to feature engineering and correlation terms on the top variables.

	feature	importance
1	debt_to_income_ratio	1208
2	credit_score	585
3	loan_amount	301
0	annual_income	216
4	interest_rate	150
20	employment_status_Retired	116
22	employment_status_Student	112
23	employment_status_Unemployed	77
6	subgrade	44
5	grade	44

Introduce 6 interaction features to capture borrower risk more precisely:

- loan to income ratio = $\frac{\text{loan amount}}{\text{annual income}}$ → How much loan is relative to income
- disposable income = annual income – loan amount → Remaining income after paying loan
- interest burden = loan amount × interest rate → high loan + high rate = exponential risk
- risk interaction = $\frac{\text{debt to income ratio}}{\text{credit score}}$ → Separates high risk profiles (high DTI ratio and low score)
- existing monthly debt = $\frac{\text{debt to income ratio} \times \text{annual income}}{12}$
- disposable monthly income = $\frac{\text{annual income}}{12} - \text{existing monthly debt}$

MODEL 2

FE + LGBM + K-FOLD CV



Outcome

Mean ROC-AUC across CV folds:
0.919692854 < Model 1

Kaggle score: 0.91964 < Model 1

	features	importances
1	num_debt_to_income_ratio	898
2	num_credit_score	530
8	num_risk_interaction	263
3	num_loan_amount	255
0	num_annual_income	137
4	num_interest_rate	125
26	cat_employment_status_Retired	122
28	cat_employment_status_Student	104
29	cat_employment_status_Unemployed	72
10	num_disposable_monthly_income	70

MODEL 3

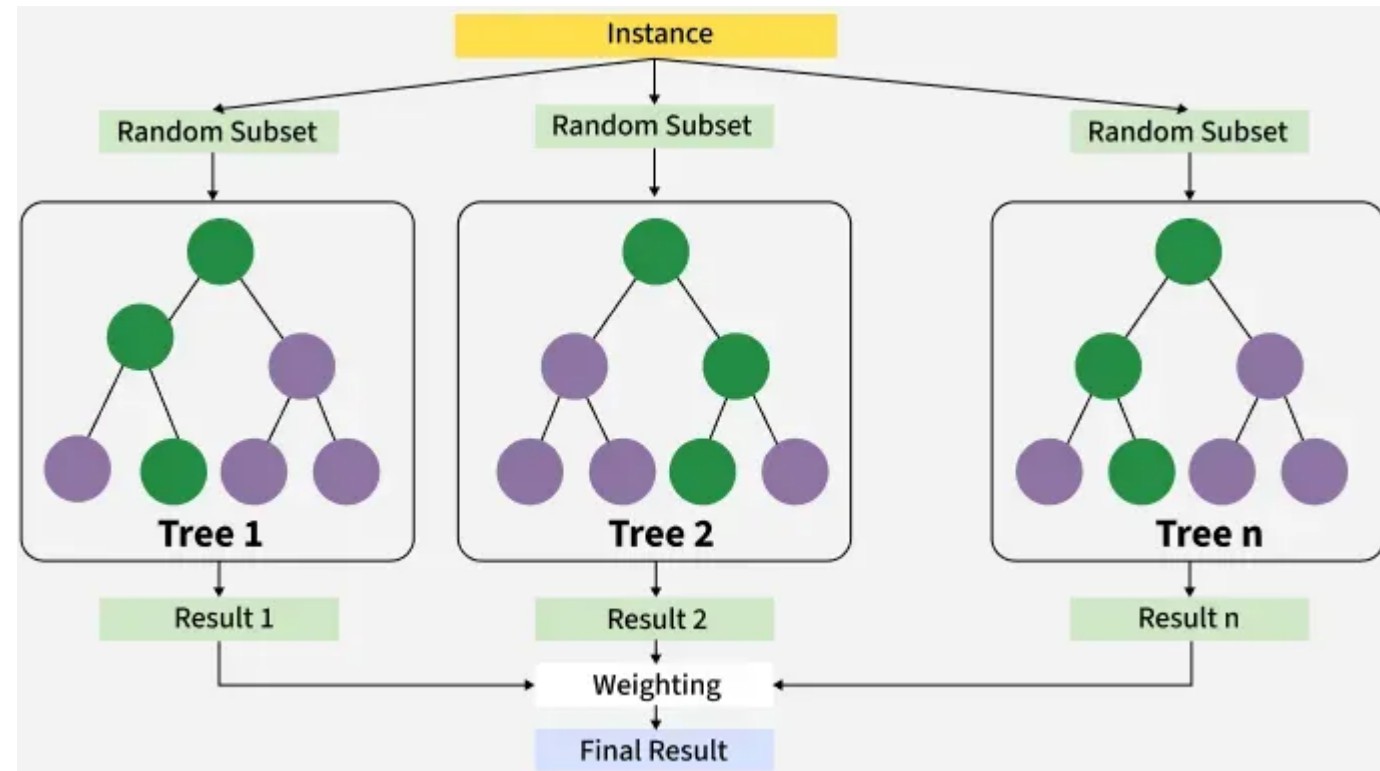
FE + XGB + K-FOLD CV



Outcome

Mean ROC-AUC across CV folds:
 $0.9201410197 > \text{Model 1}$

Kaggle score: $0.92008 > \text{Model 1}$



MODEL 4

FE + CATBOOST + K-FOLD CV



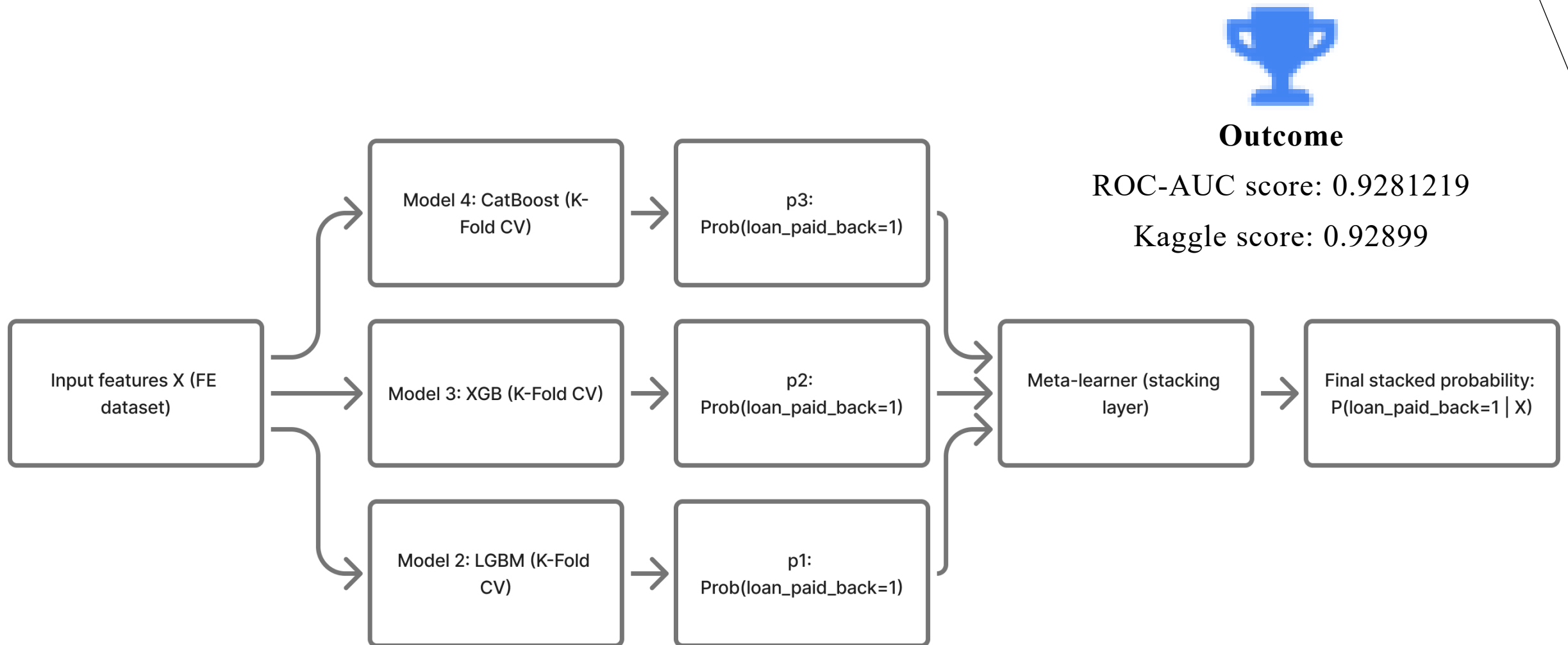
Outcome

Best ROC-AUC across CV folds:
 $0.917035 < \text{Model 1}$

Kaggle score: $0.91776 < \text{Model 1}$

	features	importances
29	cat_employment_status_Unemployed	59.234117
2	num_credit_score	9.709979
8	num_risk_interaction	7.955660
26	cat_employment_status_Retired	7.018527
1	num_debt_to_income_ratio	6.749670
28	cat_employment_status_Student	4.609124
11	num_grade	4.012560
4	num_interest_rate	0.182994
3	num_loan_amount	0.115915
10	num_disposable_monthly_income	0.113356

MODEL 5 STACKING



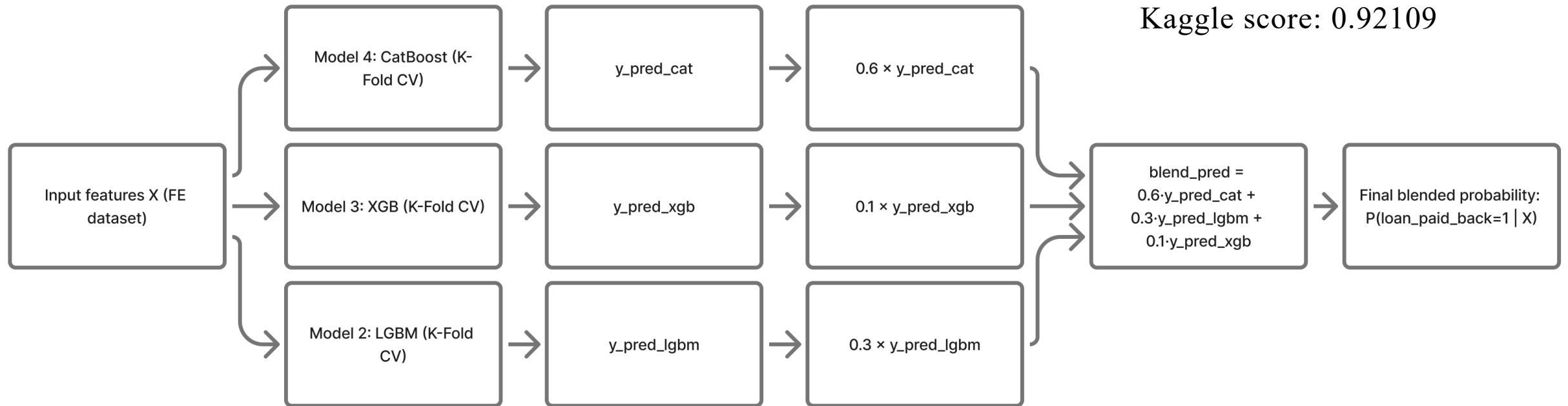
MODEL 6 BLENDING



Outcome

ROC-AUC score: 0.9268627

Kaggle score: 0.92109



SUMMARY

		CV ROC-AUC Score	Kaggle Score
Model 1 (Baseline)	LGBM Classifier + K-Fold CV	0.9198752	0.91972
Model 2	FE + LGBM Classifier + K-Fold CV	0.919692854	0.91964
Model 3	FE + XGB Classifier + K-Fold CV	0.9201410197	0.92008
Model 4	FE + CatBoost Classifier + K-Fold CV	0.917035	0.91776
Model 5	Stacking	0.9281219	0.92899
Model 6	Weighted Blending	0.9268627	0.92109

Ensemble stacking and blending models outperform the individual models.

Model 5 (stacking) achieves the highest CCV ROC-AUC and the best Kaggle score showing that improvement is not just overfitting to CV.



Predicting Loan Payback

Playground Series - Season 5, Episode 11

Playground · 3724 Teams · 7 days ago

21/3724





**THANK
YOU**